

Complete Application Training Manual

Standard Operating Procedures, Role Workflows, Material Issue/Return Lifecycle, Tool Asset Governance, and 1-Click WhatsApp Reminders.

ARCHITECTURE

**2 Independent
DBs**

OVERDUE THRESHOLD

> 7 Days Active

PLATFORM EDITION

**v2.2.3 PWA
Active**

TARGET ROLES

**Storekeepers, Supervisors, Tool In-charges,
Admins**

SYSTEM BUILD

**Enterprise Web & Mobile PWA (iOS /
Android / Desktop)**

DOCUMENT VERSION

Manual Release v2.2.3 (Current Production)

SECURITY PROTOCOL

**Role-Based Access Control + Request
Governance**

TRAINING MODULE TABLE OF CONTENTS

Module 1: System Overview & Database Separation Page 2

Module 3: Material Issue & Return Lifecycle Page 4

Module 5: Tool Asset Master Register & Governance Page 6

Module 2: User Roles & Access Control Matrix Page 3

Module 4: Overdue Management & 1-Click WhatsApp Follow-up Page 5

Module 6: Dashboard KPI Analytics & Reporting Page 7

Module 7: PWA Installation & Offline
Sync

Page
8

Module 8: Operational FAQ &
Troubleshooting

Page
9

Module 1: System Overview & Database Separation

The **CMM SMS Store Management System** is an enterprise-grade Progressive Web Application designed for end-to-end management of engineering consumables, building materials, and master tool asset registers.



CRITICAL ARCHITECTURAL RULE: Strict Database Separation

The system manages

TWO COMPLETELY INDEPENDENT AND UNCONNECTED DATABASES

in Firebase Realtime Database. Never attempt to merge, cross-link, or foreign-key records between them.

1. Material Issue Register (`issues/`)

Logs daily transactional material disbursements, quantities, and return statuses.

- **Cache:** `issuesCache`
- **Lifecycle:** Issued → Partially Returned → Returned
- **Features:** Partial return logs, camera photo receipts, overdue follow-up dialog, 1-click WhatsApp alerts.
- **Assigned Roles:** Storekeeper, Viewer, Admin

2. Tool Asset Register (`tools/`)

Master catalog of company physical capital equipment and tools.

- **Cache:** `toolsCache`
- **Serial ID:** `CMM/SMS/[TOOL]/[SEQ]`
- **Conditions:** Available, In Maintenance, Damaged, Lost (No "In Use").
- **Governance:** Deletion requests queue (`toolDeletionRequests/`). Admin approval required.
- **Assigned Roles:** Tools Admin, Tools Viewer, Admin

Module 2: User Roles & Access Control Matrix

The system enforces strict Role-Based Access Control (RBAC). When users register, they are placed in a pending state until reviewed and approved by an Administrator.

ROLE	MATERIAL ISSUES	MATERIAL RETURNS	TOOLS MASTER	TOOL DELETIONS	ADMIN PANEL
admin	Full Access	Full Access	Full Access	Direct Delete & Approvals	Full Control
storekeeper	Create & Edit	Record Returns	View Only	Request Only	No Access
tools_admin	View Only	No Access	Create & Update	Request Only	No Access
tools_viewer	No Access	No Access	Read-Only View	No Access	No Access
viewer	Read-Only View	No Access	No Access	No Access	No Access



New User Registration Workflow:

When a new team member creates an account, they select their intended role. An immediate pending access request is transmitted to the Admin Center. Administrators receive a live red notification badge and can Approve or Reject with 1 click.

Module 3: Material Issue & Return Lifecycle

1 Step 1: Issuing Material / Consumable

Navigate to **Issue Material**. Fill in Material Name, Category, Quantity Issued, and Unit (Nos, Meters, Kg, Sets). Select the Supervisor from the dropdown; their phone number auto-fills. Tag the work area/location and mark whether the item is returnable.

2 Step 2: Monitoring the Active Register

Active disbursements appear in the **Material Register**. Dynamic badges indicate current state:

• Issued (0% Returned)

• Partially Returned (1% - 99%)

• Returned (100% Complete)

3 Step 3: Recording Returns & Photo Receipts

Click **Record Return** on any active issue. Enter the quantity being returned. If less than total issued, system automatically transitions status to **Partially Returned** and recalculates pending balance.

Attach photo proof directly using mobile camera or file picker. The system automatically compresses photos to save bandwidth while maintaining receipt clarity.

Material Register Item Simulation

Partially Returned

Item: 2.5 sq mm Copper
Cable

Issued Qty: 100 Meters

Returned: 40 Meters

Pending Balance: 60
Meters

Supervisor: **Rajesh Kumar (+91 9876543210)** • Location: **Block C - 4th Floor**



Module 4: Overdue Management & 1-Click WhatsApp Follow-up

To eliminate misplaced items and minimize project losses, the system enforces an automated **7-Day Overdue Rule**. Any material issued > 7 days ago with a pending balance triggers immediate follow-up alerts across the application.

Overdue Follow-up Modal Interface (Live Simulation)

12 Days Overdue

Safety Harness Double Lanyard (Set)

Pending Qty: 2 / 2 Sets • Issued Date: 27 Jul 2026 • Area: Tower Crane 1

 Vikram Singh (Site Supervisor)

 WhatsApp

 Call



1-Click Pre-formatted WhatsApp Message Template:

Hello Vikram Singh,

This is a reminder from CMM SMS Store regarding the following overdue item:

 Material: Safety Harness Double Lanyard (Set)

 Pending Qty: 2 / 2

 Issued Date: 27/07/2026 (12 days ago)

 Area: Tower Crane 1

Please arrange for its return to the store at the earliest.

Thank you!

Module 5: Tool Asset Master Register & Governance

The **Tools Master Register** tracks all capital equipment, heavy power tools, measuring instruments, and machinery assets owned by the project.

1. Standard Serial ID Tagging Format

Every tool added receives a permanent unique serial identification code structured as:

CMM / SMS / [TOOL_CODE] / [001, 002, ...]

Examples: CMM/SMS/GRINDER/001 • CMM/SMS/DRILL/004 • CMM/SMS/TOTALSTATION/001

2. Physical Tool Condition States

CONDITION STATUS	DESCRIPTION & OPERATIONAL MEANING	ACTION REQUIRED
Available	Fully functional, calibrated, and ready in store rack.	Store ready for deployment.
In Maintenance	Under calibration, servicing, motor repair, or inspection.	Track service vendor ETA.
Damaged	Physically broken or unsafe for site operations.	Quarantine tool in scrap zone.
Lost	Unaccounted for during physical stock audit.	Trigger supervisor loss report.

3. Tool Deletion Request Workflow (Strict Governance)



Tools Cannot Be Deleted Directly by Standard Users:

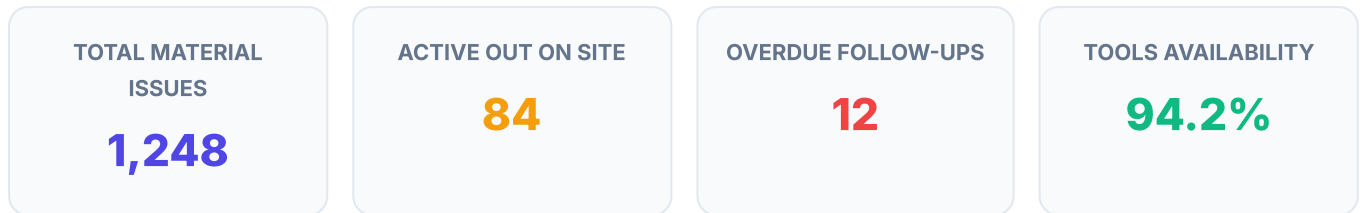
To prevent accidental loss of physical asset audit trails, non-admin users submit a

Deletion Request

with mandatory remarks. Administrators review and approve/reject deletion requests from the Admin Center.

Module 6: Dashboard KPI Analytics & Reporting

The **Live Dashboard** provides immediate visibility into store operations, material turnover rate, tool asset health, and stock risks.



One-Click Excel / CSV Export & Audit Backups

Export full transaction ledgers with 1 click for management reviews and billing reconciliations:

- **Material Issue Ledger Export:** Includes Issue ID, Timestamp, Supervisor, Material, Qty Issued, Qty Returned, Balance, Area, and Photo Links.
- **Tools Asset Inventory Export:** Includes Serial Tag, Category, Shelf Location, Condition, Purchase Date, and Audit Log count.

📦 Module 7: PWA Installation & Offline Sync

CMM SMS Store functions as an installable **Progressive Web App (PWA)** on iOS iPhones/iPads, Android devices, and Windows/Mac desktops without needing App Store downloads.

📱 iOS Safari Installation

1. Open Safari and load the store URL.
2. Tap the **Share** button (box with upward arrow).
3. Scroll down and tap **Add to Home Screen**.
4. Tap **Add**. Launch from your home screen.

🖥️ Android & Desktop Installation

1. Open in Chrome or Edge.
2. Click the **Install App** prompt in the address bar or bottom banner.
3. Confirm install. The app opens in dedicated full-screen window mode.

Offline Resilience & Realtime Sync

The app features an integrated Service Worker engine (`sw.js`) with client-side caching. If site internet drops, the app remains responsive, displaying an amber "Working Offline" status bar. When connectivity returns, pending records sync automatically with Firebase Realtime Database.

Module 8: Operational FAQ & Troubleshooting

Frequently Asked Questions

Q: Why doesn't the WhatsApp reminder open the contact on desktop?

A: On desktop browsers, clicking WhatsApp directs to **WhatsApp Web** (web.whatsapp.com). Ensure you have scanned your WhatsApp Web QR code once. On mobile devices, it launches the native WhatsApp application directly.

Q: How do I change a supervisor's phone number after issue submission?

A: Storekeepers and Admins can click **Edit Issue** from the Material Register table to update contact numbers, work areas, or notes at any time.

Q: A tool was scrapped. How do I delete it from the master catalog?

A: First change its status to **Damaged** with remarks. Next, click **Request Deletion**. An Administrator will review the scrap justification and execute final deletion from the Admin Center.



Technical Support & System Administration:

For user access approvals, role escalations, or database backups, contact your site System Administrator or Store Manager.